ZATIARAS JUICE

Software Architecture & Technical Design

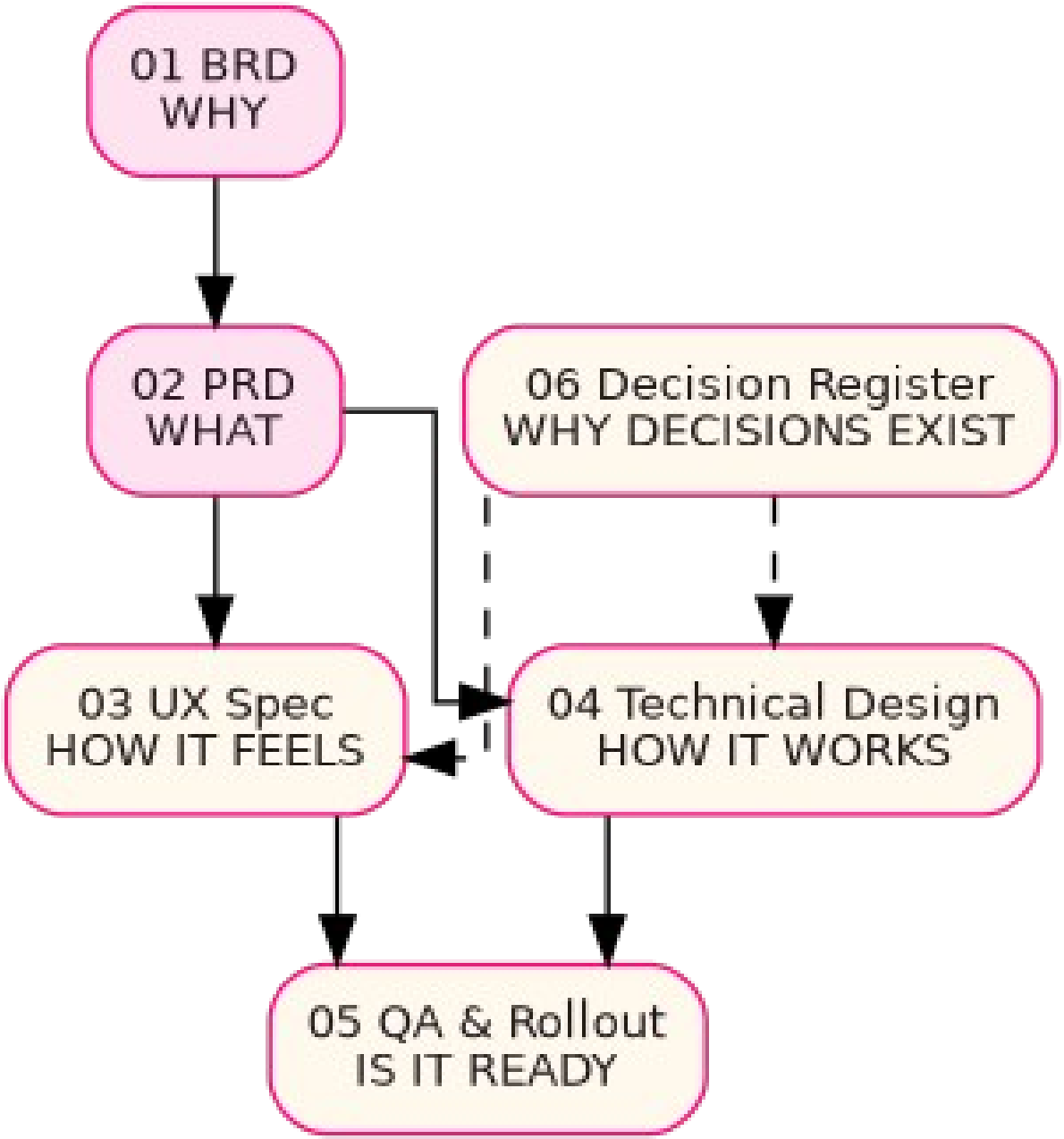
Zatiaras Web V1 - System boundaries, stack, data, integration and delivery architecture

Document code	ZJW-TECH-004
Version	1.1
Status	Reconciled Baseline
Planning cut-off	15 September 2026
Agent companion	docs/ARCHITECTURE.md + docs/CONTRACTS.md
Normative basis	Reconciled domain-specific source hierarchy; 3 official brand PDFs govern visual identity; legacy digital BRD excluded

Purpose

This document is part of the reconciled Zatiaras Web V1 source-of-truth set for human review and controlled handoff to AI coding agents. It is synchronized with a repo-native companion file and the official brand PDF authority set. The legacy digital BRD is not normative.

Document set, source authority and companion mapping



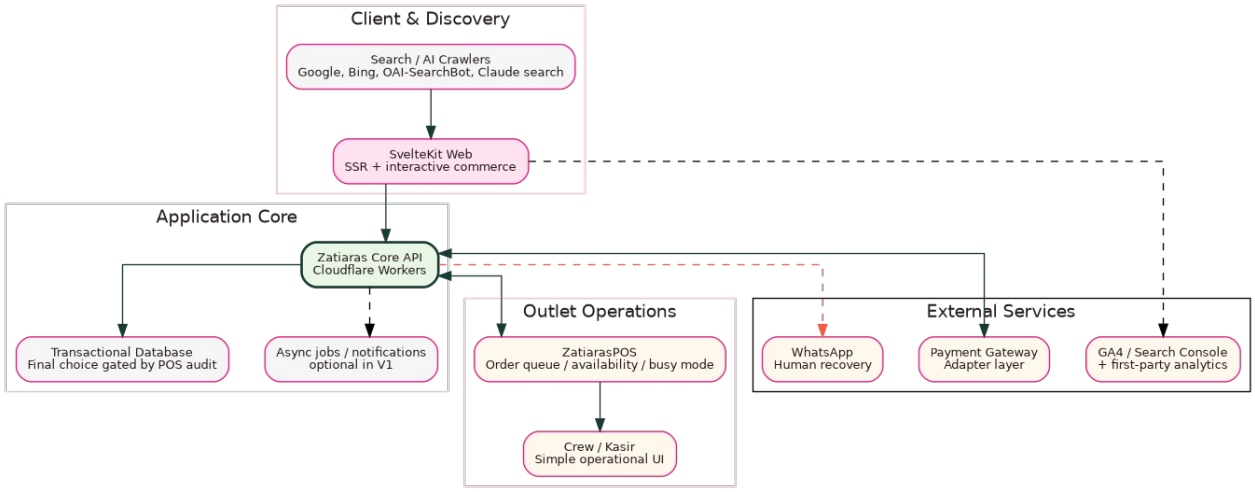
Human specification	Repo-native companion	Primary authority
01 BRD	docs/PRODUCT.md	Business intent, outcomes, scope
02 PRD	docs/SPEC_V1.md	Product behavior and business rules
03 UX Specification	docs/UX.md	Interaction and user journeys
04 Technical Design	docs/ARCHITECTURE.md + docs/CONTRACTS.md	Architecture, boundaries and contracts
05 QA / Rollout	docs/QUALITY.md	Definition of correct and launch gates
06 Decision Register	AGENTS.md + referenced decision IDs	Decision rationale and revisit triggers
3 official Brand PDFs	docs/BRAND_REFERENCES.md	Visual identity and application authority

Current document mapping
04 Software Architecture & Technical Design is paired with docs/ARCHITECTURE.md + docs/CONTRACTS.md. If prose conflicts with the official brand PDFs on visual identity, the brand PDFs win for that domain. If implementation conflicts with approved requirements, the conflict must be surfaced rather than guessed.

1. Architecture goals

- Fast mobile commerce and server-rendered public discovery.
- One shared business rule layer between customer web and ZatiarasPOS where practical.
- Payment correctness and idempotent state transitions.
- Operational availability rather than false omnichannel inventory precision.
- Human recovery for payment/fulfillment exceptions.
- Good observability and testability for AI-agent-assisted development.
- Progressive architecture: ready for later kiosk/CRM/loyalty without overengineering V1.

2. System context



3. Recommended technology stack

Layer	Recommended choice	Rationale / status
Customer frontend	SvelteKit + TypeScript	Strong SSR + interactive app model; suitable for fast mobile commerce
Runtime / hosting	Cloudflare Workers	Cloudflare documents first-class SvelteKit deployment on Workers [EXT-7]
Motion	CSS/Svelte transitions + GSAP/ScrollTrigger selectively	Use advanced motion only on brand/discovery surfaces
Application core	Zatiaras Core API on Workers	Centralizes order/payment/availability/support rules
Database	Decision gated by ZatiarasPOS audit	Prefer one transactional source of truth; avoid dual databases without reason
Media	Cloudflare R2 if dynamic uploads are required	Useful for complaint images/media; static product assets may ship with app

Layer	Recommended choice	Rationale / status
Payment	Provider adapter layer	Allow one production provider while avoiding hard lock-in
Analytics	GA4/Search Console + first-party events	Marketing attribution separate from commerce truth
Human support	WhatsApp click-to-chat / Business integration	Trust layer for exceptions
Testing	Vitest + Playwright (or equivalent)	Fast unit/integration plus real user-flow E2E

4. Rendering, SEO and AI discovery architecture

Concern	Technical requirement
Public pages	SSR or prerendered HTML with meaningful text before hydration
Structured data	Organization/LocalBusiness/Breadcrumb; Product/Offer where accurate and appropriate [EXT-2]
Local entity	Branch-specific stable URLs + consistent address/hours/phone/menu facts
Google AI features	Follow normal SEO; local/product/business information remains relevant [EXT-3]
ChatGPT discovery	Do not block OAI-SearchBot if discovery/citation is desired [EXT-4]
Claude discovery	Allow Claude-SearchBot/Claude-User if desired; training crawler policy is separate [EXT-5]
Sitemap/indexing	Generate sitemap; manage canonical/noindex for transactional/private routes
Performance	Image optimization, code splitting, minimal client JS on public pages

5. Domain model

Entity	Purpose	Key relationships
Outlet	Branch operating data	Has menu availability, busy mode, pickup rules
MenuItem	Product master	Has variants/options; shown across outlets
OutletMenuAvailability	Operational sellability	Outlet + item + status; not exact inventory
Order	Customer transaction intent	Belongs to outlet; contains items and lifecycle status
OrderItem	Snapshot of purchased item/price/options	Belongs to order

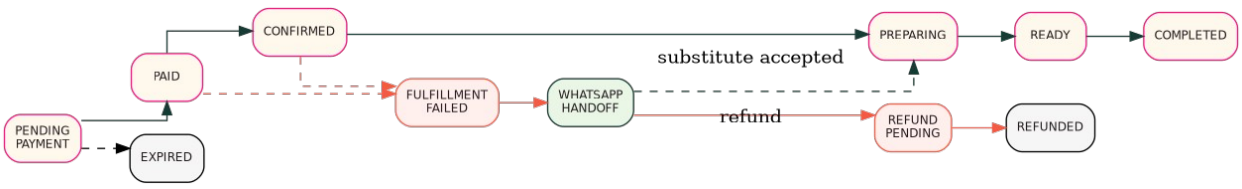
Entity	Purpose	Key relationships
Payment	Gateway transaction record	Belongs to order; webhook events idempotent
Refund	Money return lifecycle	Belongs to payment/order
SupportCase	Exception or complaint record	May link to order/customer
OrderEvent	Audit timeline	Immutable/append-style state history recommended
PickupSlot	Operational capacity window	Outlet/date/time; optional configurable capacity

6. Core API boundary

Operation	Purpose
getOutlets / getMenu	Public discovery and ordering data
createOrder	Validate cart/outlet/availability and create pending order
createPayment	Create payment request for valid pending order
handlePaymentWebhook	Verify provider signature, idempotently update payment/order
markPreparing / markReady / completeOrder	Outlet lifecycle controls
failFulfillment	Record inability to fulfill and open recovery state
requestRefund / markRefunded	Controlled money-recovery lifecycle
setAvailability / setBusyMode	Outlet operational control
createSupportCase	Record complaint/exception context

Agentic coding rule
Generate or maintain executable contracts (TypeScript/Zod/OpenAPI/tests) from the approved human specification. If executable contracts conflict with approved product requirements, flag the conflict rather than silently choosing.

7. Order and payment lifecycle



Technical invariant	Implementation expectation
Browser cannot mark payment successful	Only verified server-side provider callback/status may do so
Webhook idempotency	Store provider event/transaction identifiers; duplicate delivery has no duplicate side effects

Technical invariant	Implementation expectation
Money vs fulfillment separate	PAID does not mean PREPARING; failure after payment has explicit state
Auditability	Important state changes record actor/time/reason
Group order approval	Large request reaches approval before payment creation

8. Availability and cross-channel model

GoFood, GrabFood, ShopeeFood and walk-in demand are not fully integrated with ZatiarasPOS. Therefore V1 must not expose an exact real-time inventory promise. Use operational availability plus outlet load controls.

Control	Values / behavior
Menu availability	AVAILABLE / LIMITED / SOLD_OUT
Outlet load mode	NORMAL / BUSY / VERY_BUSY / PAUSE_WEB_ORDER
Pickup slots	Can lengthen/limit order promise during load
Manual override	Crew may correct operational reality immediately
Future integration	Exact ingredient/BOM inventory only after reliable omnichannel data exists

9. ZatiarasPOS integration

- Web orders should appear in the existing operational environment rather than requiring staff to monitor a second complex dashboard.
- POS and web should share order status definitions and critical domain contracts, but customer and staff UI components need not be shared.
- Existing POS database/backend must be audited before locking D1 vs PostgreSQL or creating a new persistence layer.
- External marketplace orders remain parallel V1 channels; end-of-shift reconciliation may be added without pretending real-time integration.

10. Security, privacy and reliability

Area	Requirement
Secrets	Payment/API credentials stored server-side only
Authorization	Crew/admin actions protected by role/session; no public order mutation endpoint
Rate limiting	Protect order creation, lookup, feedback and auth endpoints
PII minimization	Collect only data needed for ordering/support; avoid excessive personal data
Order tracking access	Use secure unguessable token/identifier or equivalent protection
Payment validation	Verify amount/order/provider metadata, not just “success” string
Logging	Structured events without leaking secrets or unnecessary

Area	Requirement
	customer data
Backups/recovery	Database backup/recovery capability documented before launch

11. Observability and analytics

Event family	Examples
Commerce	order_created, payment_created, payment_succeeded, order_confirmed, order_ready, order_completed
Failure	payment_expired, fulfillment_failed, refund_requested, refund_completed
Funnel	product_view, add_to_cart, checkout_started
Support	whatsapp_support_clicked, support_case_created
Discovery	source/UTM, Google organic, Maps, Instagram, AI referrals where detectable
Operations	prep duration, busy mode, availability changes

12. Agentic engineering delivery model

The project is intended for AI-agent-assisted implementation. Human PDF specifications remain governance artifacts; repo-native instructions and executable contracts should be derived from them. Modern coding agents can consume AGENTS.md/repository instructions, but tests and schemas must remain the ultimate machine-checkable evaluator. [EXT-10][EXT-11]

Repo artifact	Role
AGENTS.md	Stable cross-agent working rules and project navigation
docs/PRODUCT.md + SPEC_V1.md	Machine-friendly summary of approved product behavior
OpenAPI / schemas / TypeScript types	Executable contracts
tests/	Behavioral evaluator for agent output
tasks/	Small implementation slices with acceptance criteria
ADR files	Technical decision rationale and revisit triggers

13. Open technical decisions

Decision	Current status	Resolution gate
Transactional database: D1 vs PostgreSQL/current POS store	OPEN	Audit ZatiarasPOS data layer/schema and deployment
Production payment provider	OPEN	Commercial/pricing/refund/webhook confirmation at implementation time
Exact group-order threshold	OPEN	Outlet capacity test and operating

Decision	Current status	Resolution gate
		policy
Pickup-slot capacity algorithm	OPEN	Pilot observation; begin simple/configurable
Notification automation beyond web status	DEFER	Add only if operational value proven

Sources and evidence basis

Source authority is domain-specific. Business/product intent is governed by the reconciled BRD, PRD and Decision Register; UX by the UX Specification; technical implementation by Technical Design plus executable contracts/tests; visual identity by the three official brand PDFs (Asset Master Pack, Brand Identity Guidelines, Application & Rollout Kit). Internal operating data supports evidence, external official guidance supports implementation/research, and the legacy digital BRD is not normative.

Internal sources

[INT-0] Zatiaras Web V1 reconciled source-of-truth model. Business intent is governed by the current BRD/PRD/Decision Register; UX by the UX Specification; technical implementation by Technical Design plus executable contracts/tests; brand/visual identity by the three official brand PDFs listed below. Project conversation is provenance and working context, not authority over approved brand assets. The legacy digital BRD is intentionally excluded as a normative source.

[INT-1] Project planning conversation and research through 15 September 2026. Used as provenance for decisions captured in this reconciled specification set; not a substitute for approved source documents.

[INT-2A] Zatiaras Brand Asset Master Pack V1.0, 12 pages. Highest brand authority for exact assets, logo hierarchy/protection, approved variants, graphic-device assets, production rules, cup branding, vendor handoff and brand QA.

[INT-2B] Zatiaras Juice Brand Identity Guidelines V1.2 Reconciled, 12 pages. Brand authority for essence, messaging hierarchy, color system, typography, slogan system, graphic language, product imagery and digital identity behavior.

[INT-2C] Zatiaras Juice Application & Rollout Kit V1.0, 12 pages. Brand authority for touchpoint/application patterns including social, marketplaces, GBP, menu/price communication, signage, packaging, campaign templates and branch rollout.

[INT-3] POS_TRANSAKSI_SAMARINDA_2025-11-26_sd_2026-06-26.xlsx. Historical baseline only; not current September 2026 performance and does not reliably separate Juanda vs Slamet Riyadi.

[INT-4] MASTER_HPP_SEMUA_MENU_ZATIARAS_2026-06-30.xlsx and MASTER_SOP_PORSI_ZATIARAS_SEMUA_CABANG_2026-06-29.xlsx. Cost/portion context; current supplier, shrinkage and OPEX still require operational reconciliation.

[INT-5] MASTER_HARGA_MENU_ZATIARAS_3_KOTA_AS_OF_2026-06-29.xlsx. Historical price master; verify current prices before production publication.

External research and official guidance

[EXT-1] Google Business Profile Help, “Cara memperbaiki peringkat lokal di Google”. Local results are mainly based on relevance, distance and prominence. https://support.google.com/business/answer/7091?hl=id

[EXT-2] Google Search Central, LocalBusiness structured data. https://developers.google.com/search/docs/appearance/structured-data/local-business?hl=id

[EXT-3] Google Search Central, “Mengoptimalkan situs Anda untuk fitur AI generatif di Google Penelusuran” (2026). SEO remains foundational; local, product, image and business information matter. https://developers.google.com/search/docs/fundamentals/ai-optimization-guide?hl=id

[EXT-4] OpenAI Help Center, Publishers and Developers FAQ. Public sites can appear in ChatGPT search; OAI-SearchBot access supports discovery, citation and linking. https://help.openai.com/id-id/articles/12627856-penerbit-dan-pengembang-faq

[EXT-5] Anthropic Help Center, web crawler controls. Claude-User and Claude-SearchBot support user-directed retrieval and search indexing. https://support.anthropic.com/en/articles/8896518-does-anthropic-crawl-data-from-the-web-and-how-can-site-owners-block-the-crawler

[EXT-6] web.dev, Core Web Vitals. Recommended “good” thresholds at p75: LCP <= 2.5 s, INP <= 200 ms, CLS <= 0.1. https://web.dev/articles/vitals

[EXT-7] Cloudflare Workers docs, SvelteKit. SvelteKit is supported for full-stack deployment on Workers. https://developers.cloudflare.com/workers/framework-guides/web-apps/sveltekit/

[EXT-8] Open Data Pemerintah Kota Samarinda, UMKM by kecamatan and type, 2023-2025, updated 19 June 2026. https://opendata.samarindakota.go.id/

[EXT-9] Portal Resmi Kota Samarinda, city statistics/transparency. Population about 893,385; 2025 PDRB about Rp106.62T and economic growth 6.22%. https://samarindakota.go.id/

[EXT-10] GitHub Docs, repository and agent instructions (AGENTS.md and repository custom instructions) for coding agents. https://docs.github.com/en/copilot/how-tos/configure-custom-instructions-in-your-ide/add-repository-instructions-in-your-ide

[EXT-11] OpenAI, “Unrolling the Codex agent loop” (2026), describing repository-level AGENTS.md context discovery. https://openai.com/index/unrolling-the-codex-agent-loop/